APPENDIX A

Figures

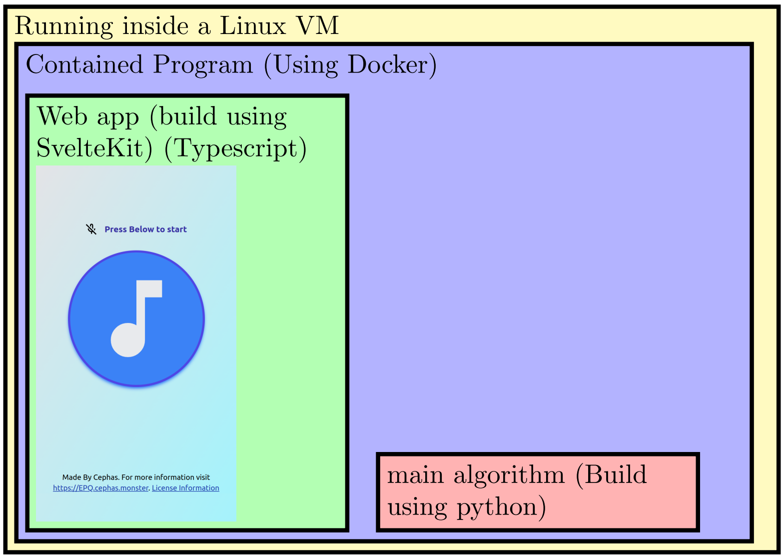


Figure A.1: Overall Architecture

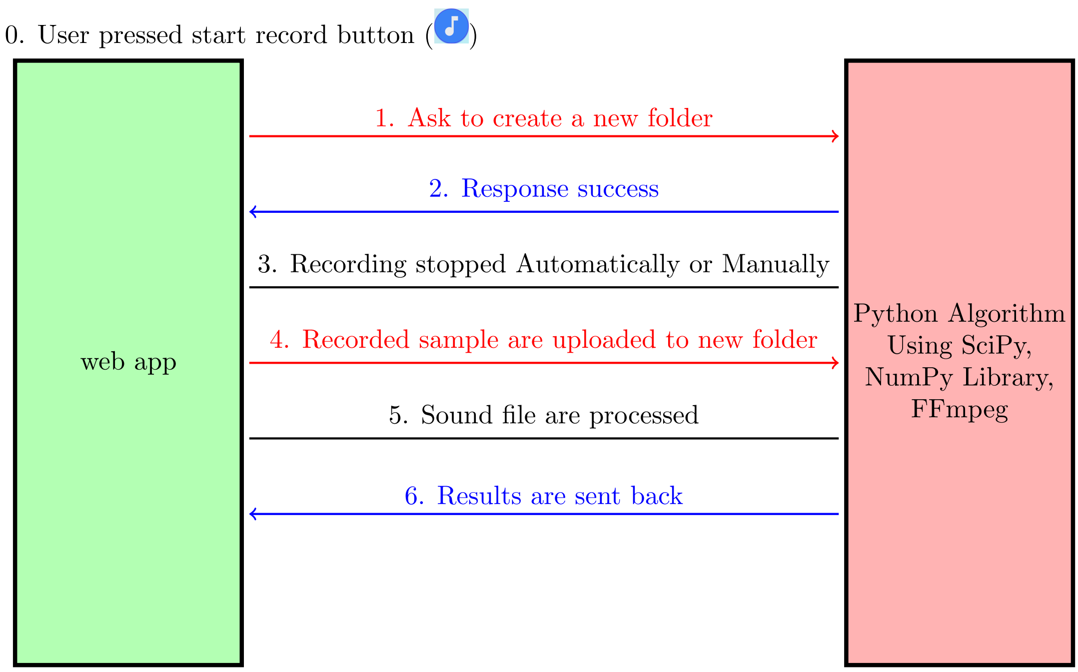


Figure A.2: Overall Architecture (app)

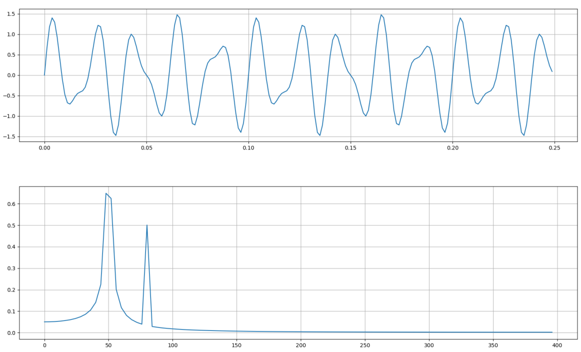


Figure A.3: Fast Fourier Transform as an example through transforming the top signal in to representation in the frequency domain at the bottom.

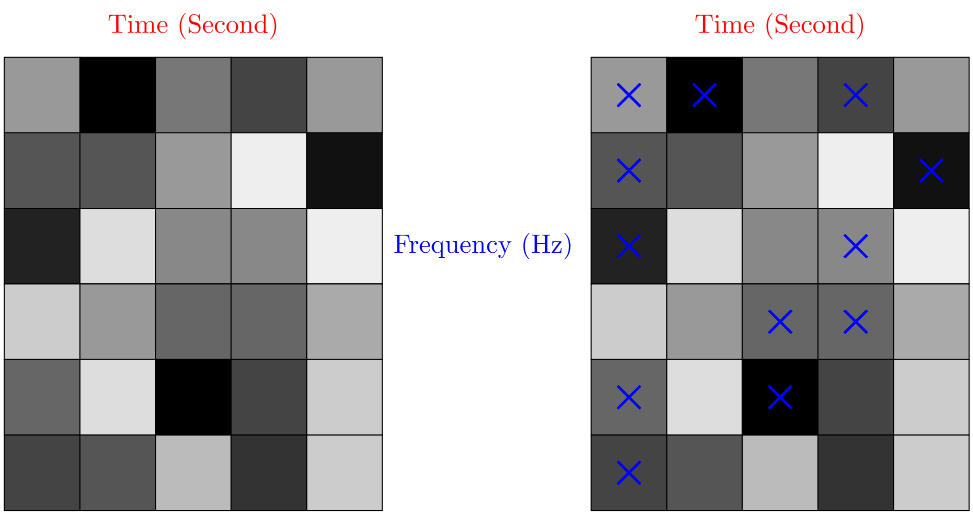


Figure A.4: Illustrating an example of a spectrogram picking of points. The darker the box, the higher the amplitude (value) represents

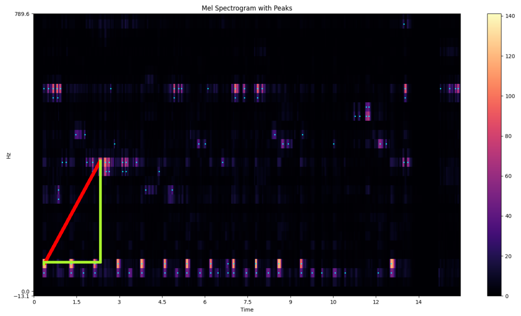


Figure A.5: The tip of the triangle represents Point A and Point B. Y-axis represents the time, X-axis represents the frequency, The amplitude scales are represented by the colour bar on the right.

Point A (Hz)	Point B (Hz)	Time delta (s)	Point A time (s)	Track UUID
20.1	302.1	1.8	0.3	*ea47a3c5*
2388532207457024332			2.1	*ea47a3c5*

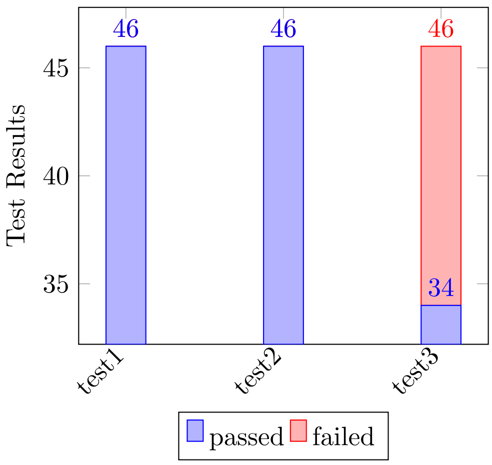
Figure A.6: Demonstrate / example of how item are hashed.

Song_ID	Name	Author	Genre
f633c47a	J. S. Bach: Prelude in C - BWV 846	Kevin MacLeod	Classical

Figure A.7: Example of a database entry

APPENDIX B

Testing results



Test 1: (tested on 100% of the song library)

filepath	passed	SUBTOTAL
test/test_1.py	46	46
TOTAL	46	46

Test 2: (tested on 100% of the song library)

filepath	passed	SUBTOTAL
test/test_2.py	46	46
TOTAL	46	46

Test 3: (tested on 100% of the song library)

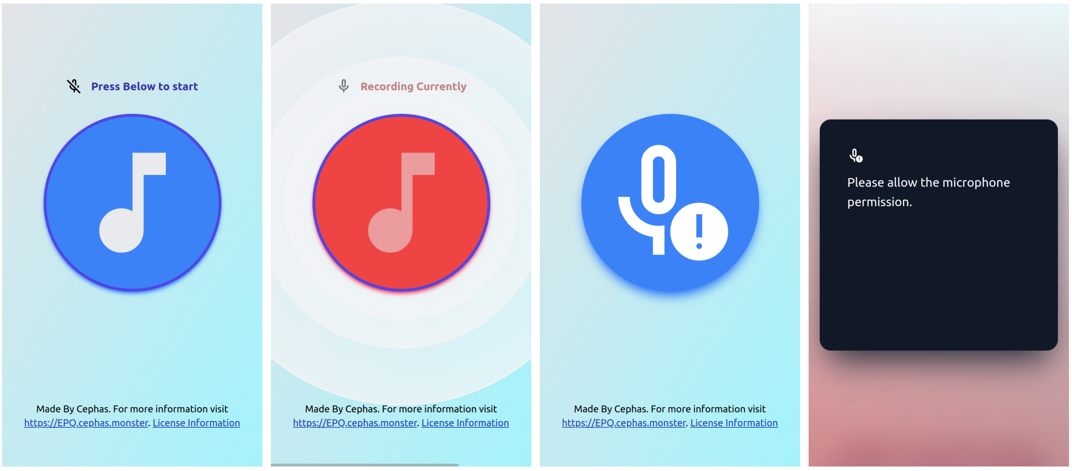
filepath	passed	failed	SUBTOTAL
test/test_3.py	34	12	46
TOTAL	34	12	46

Test 4: Most works (Only tested 40% of the song library)
Test 5: Most failed (Only tested 20% of the song library)
Test 6: All failed (Only tested 10% of the song library)

Figure B.1: test result table

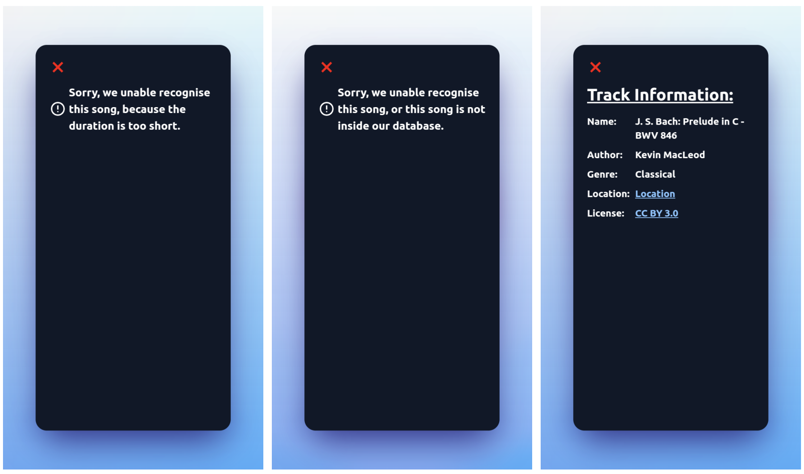
APPENDIX C

Screenshots



(a) (b) (c) (d)

Figure C.1: (a) "Home", (b) Recording (c)(d) "Microphone permission are disabled"



(a) (b) (c)

Figure C.2: (a) "Too Short", (b) "No song", (c) "Recognized song (e.g: Prelude by Bach)"

All the figures are simulated to the size of an iPhone® SE display.

APPENDIX D

Meeting logs

D.1 Meeting with the supervisor at 25 November 2024

Summary:

Improve on the part A of the production logs, with more details on the titles described and what resources I will be using (such as the URLs should be provided), the timescale and what is involve.

As things I should be including to create a great project is the benefit that this might bring to the society and the impact.

And a plan to see what I should be sending or showing to the examiner in the future such as:

- Send product log
- Presentation of the artefact
- Evidence of the artefact
- Critical bibliography
- Place that in the Appendix and write a reference back to the appendix through the production log.

D.2 Meeting with a professional at 13 January 2025

Summary:

I explained the background details during the meeting.

I have explained that the program can only detect sounds from following test that I conducted such as:

- A test where there are no modifications to the original sound,
- A test where the song has cropped and reduced to a smaller size.
- A test where there is a constant noise generated by a program with a constant sine wave in the background.
- A test where the songs are mixed with a normal acceptable background noise such as coughing.

All of those above works as expected.

However, the test that I did by recording when the sound is played on a music player on a different device doesn't work.

As an additional note, he has mentioned that the program is slow because of a reason which might be caused by the hash point not being an integer, as the hash value using integer is faster than using string as hash. (It was diagnosed with a slow database reading speed prior to the meeting).

The project has concluded as partially successful from the results produced from the searching algorithm.

Summary of Improvement I need to make from the project.

I should use a different hashing algorithm for the fingerprinting part of the program.

Round up the frequency, and the changes in time that are fed into the hashing algorithm.

Additional note:

The meetings are conducted with the consent of the professional. (in regards to the ethical issues)

The professional is my computer science teacher, as he has a decent qualification of a university degree.

APPENDIX E

Critical Bibliography

Sources		Critics	Notes
Title: Short-time Fourier transform and superoscillations Author: Alpay et al. Issue Date: November 2024 URL: https://www.sciencedirect.com/science/article/pii/S1063520324000666	Alpay et al. 2024	This article is indented to provide information about the basics of Short-time Fourier Transform.	
Title: *What is MIDI?* Author: Anthony Issue Date: February 2025 URL: https://www.gear4music.com/blog/what-is-midi/	Anthony 2025	This tutorial article is indented to provide a really basic definition about the MIDI file.	
Title: *Grokking algorithms* Author: Bhargava Issue Date: 2016 URL:	Bhargava 2016	This book is indented to provide a really basic definition about some algorithm that is involve in this project.	
Title: Query by humming Author: Ghias et al. Issue Date: 1995 URL: http://portal.acm.org/citation.cfm?doid=217279.215273	Ghias et al. 1995	This conference article provide a discussion point on how to make the artefact if I want to do it differently. This source is credible to an extent as it's written by ACM which is a scientific and educational computing society. However, it might be slightly outdated (published date 1995) as there might be more information about this.	
Title: Array programming with NumPy Author: Harris et al. Issue Date: September 2020 URL: https://www.nature.com/articles/s41586-020-2649-2	Harris et al. 2020	This program is indented to be used in the program. This program extensively used in data analysis. It's an industry standard tool to analysis array in all fields, such as research.	Utilized only in the code

Title: *100,000 tracks being uploaded to Spotify and other services daily* Author: Ingham Issue Date: October 2022 URL: https://www.musicbusinessworldwide.com/its-happened-100000-tracks-are-now-being-uploaded/	Ingham 2022	This article provide insight into the music industry from the former CEO of a large record label company Warner Bros explaining the number of music being uploaded daily.	
Title: *Abracadabra: How does Shazam work?* Author: MacLeod Issue Date: February 2022 URL: https://www.cameronmacleod.com/blog/how-does-shazam-work	MacLeod 2022	This blog post is important because this includes much more details that the original article from the Shazam company. However, this does not mean this blog post are accurate as this is from written from a perspective of the original creator. But the writer is partially qualified to write about this topic as the writer work as a software engineer at Bloomberg and a graduate of University of Edinburgh.	
Title: *INTERVIEW: Dr Avery Wang, Principal Research Scientist at Apple and Co-Founder and Chief Scientist, Shazam* Author: Newnham Issue Date: February 2023 URL: https://daniellenewnham.medium.com/fostering-innovation-and-choosing-the-right-founding-team-6287f7eed9f	Newnham 2023	This interview give an insight into the creator of Shazam Avery Wang's identity, which his source is used as concrete source in the program. Wang and III 2013; Wang 2003	

Title: Sound Spectrum Influences Auditory Distance Perception of Sound Sources Located in a Room Environment Author: Spiousas et al. Issue Date: June 2017 URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5479918/	Spiousas et al. 2017	This article provides evidence to show that for sound play from a distance, the frequency are propagated and changed. This factor is consider in the project.	
Title: *The Fast Fourier Transform Algorithm* Author: Brunton Issue Date: April 2020 URL: https://www.youtube.com/watch?v=toj_IoCQE-4	Brunton 2020	This video tutorial from the university of washington gives a basic detail and tutorial on the topic of fast fourier transform. This gives enough insight to allow me to create the project.	
Title: SciPy 1.0 Author: Virtanen et al. Issue Date: March 2020 URL: https://www.nature.com/articles/s41592-019-0686-2	Virtanen et al. 2020	This program is indented to be used in the program. This program extensively used in solving mathematical problem and scientific tool. And it's an industry standard programming interface for solving mathematical problem and analysing scientific problem.	Utilized only in the code
Title: An Industrial-Strength Audio Search Algorithm Author: Wang Issue Date: January 2003 URL:	Wang 2003	This is the original article explaining from a prospective of the creator. This article creates a basic understanding of how the basics works. They have a detail analysis of recognition rate and the performance associated. However, the article is missing from explaining the method of how they create Spectrogram, and the algorithm they are using to create it. There might be changes to the modern day algorithm that the product is using, this might be due to the age of this paper being published in 2003. And since now the company are acquired by Apple Inc.	

Title: Systems and methods for recognizing sound and music signals in high noise and distortion Author: Wang and III Issue Date: February 2013 URL: https://patents.google.com/patent/US8386258B2/en?assignee=Shazam+Investments+Ltd&num=25	Wang and III 2013	This source is very credible and concrete, as the original program Shazam are based on it. It is created by the founder and principle scientist of Shazam. Inside the source there are also figures attached to it. This patent is cited (as of February 2025) 771 times. It's being used in various different field, most notability social media companies and big tech companies such as Google, Snap. However, the source might not be as relevant because the patent are outdated and expired.	
Title: *Music Database Retrieval Based on Spectral Similarity* Author: Yang Issue Date: 2001 URL: http://ilpubs.stanford.edu:8090/489/	Yang 2001	This source provides resources mainly how to test the artefact. This report are also written by the Stanford University which is a very credible source and provides great information.	

APPENDIX F

Timetable

See the landscape table file after this page.

- Green done means on time
- Orange done means not on time

I choose to use timetable target set by week because it allows for more streamline approach to time management.

timetable

Todos	Expected Length	Recommanded Date	Final Length	Final Date	Done	Notes
learning about how spectrogram are made	1 week	27/10/2024	1 week	5/11/2024	✓	/
Make a spectrogram	1 week	4/11/2024	2 week	20/11/2024	✓	The creation process takes longer than expected, as I have to figure out out of memory issues.
make a contellation map of it	2 week	18/11/2024	1 week	1/12/2024	✓	
store the data point as a hash point	2 week	2/12/2024	2 week	2/12/2024	✓	The Time it takes to complete it are shorter than expected
place the hash point in a database	2 week	16/12/2024	3 week	25/12/2024	✓	The process took longer than expected due to unexpected event happening outside of the EPQ.
make a web app that allows recording of the music	2 week	16/1/2025	2 week	30/1/2025	✓	
linking everything together	2 week	1/2/2025	3 week	6/2/2025	✓	Both process are interlinked to each other and the time of completion are linked
testing process	2 week	13/2/2025	1 week	13/2/2025	✓	
write a report about it	3 week	13/3/2025	1 week	13/3/2025	✓	Both process are interlinked to each other as I am required to test the program while writing the report
make slideshow to present	2 week	13/3/2025	1 week	13/3/2025	✓	

APPENDIX G

Links

The Artefact: https://recognizer.cephas.monster/	
EPQ directory: https://EPQ.cephas.monster/	
Privacy statement: https://EPQ.cephas.monster/privacy	
EPQ code: https://EPQ.cephas.monster/code	

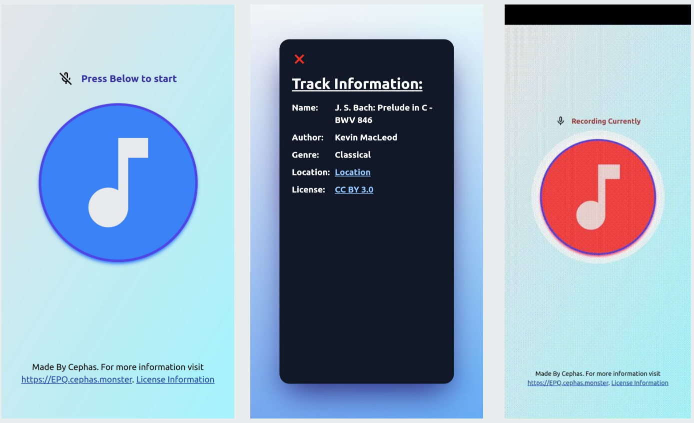
This report: https://EPQ.cephas.monster/report	

APPENDIX H

Presentation

To explore the mechanism that allows music recognition apps to work.

By Cephas



QR codes for all the links

All the links

The artefact

Question

How well can you recognise a song?

I am certainly not good at it. 😔

But how can computers recognise songs better than us?

Why did I choose this title

To explore the mechanism that allows music recognition apps to work.

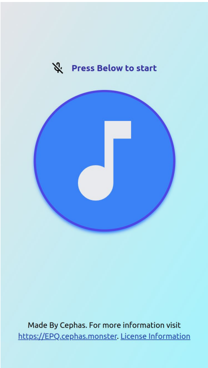
- Interesting
- It has purpose
- Learning new things
- Difficult / Challenging
- Problem-solving
- Rewarding

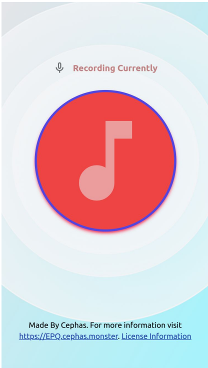
Source and resources I used

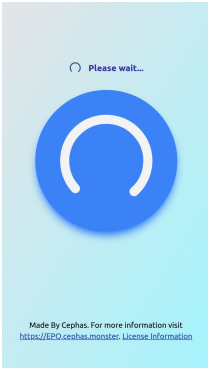
- Patent from the original creator of Shazam
- Research article
  - How is it made
  - Research about the testing method
- Blog post made by a professional
- Thanks to my computer science teacher listening for feedback.

An oversimplied summary of how it works

The user records the song	The song is uploaded to the server	The song is processed	The key feature of the song are picked out







An oversimplied summary of how it works

pick out key feature of the song

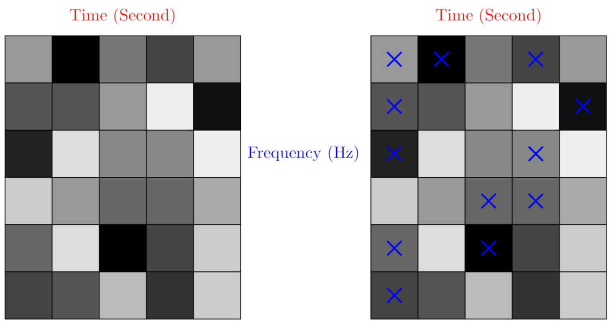
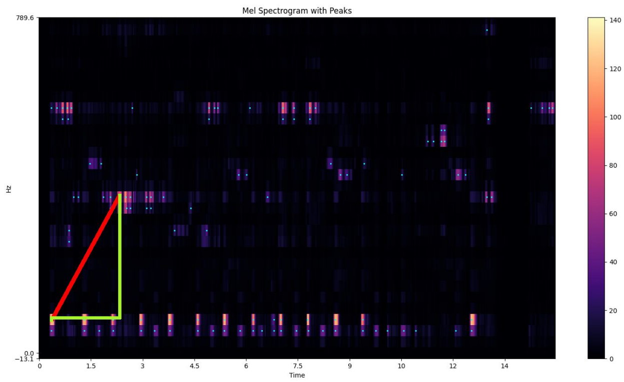


Figure A.4: Illustrating an example of a spectrogram picking of points. The darker the box, the higher the amplitude (value) represents

An oversimplied summary of how it works

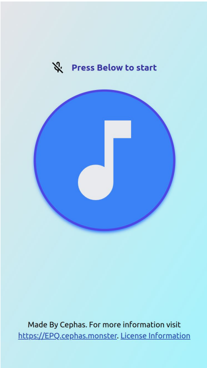
pick out key feature of the song

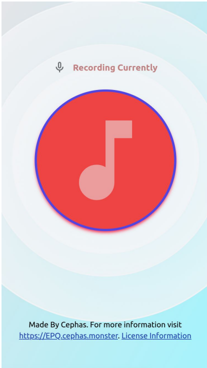


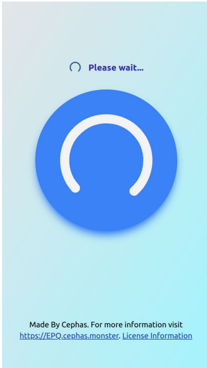
Point A (Hz)	Point B (Hz)	Time delta (s)	Point A time (s)	Track UUID
20.1	302.1	1.8	0.3	*ea47a3c5*

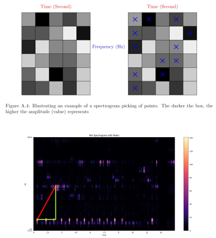
An oversimplied summary of how it works

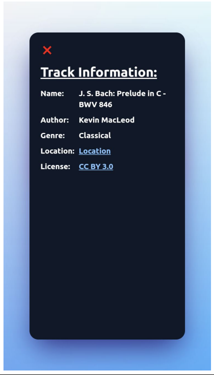
The user records the song	The song are uploaded to the server	The song are processed	The key feature of the song are picked out	The features are compared with the database

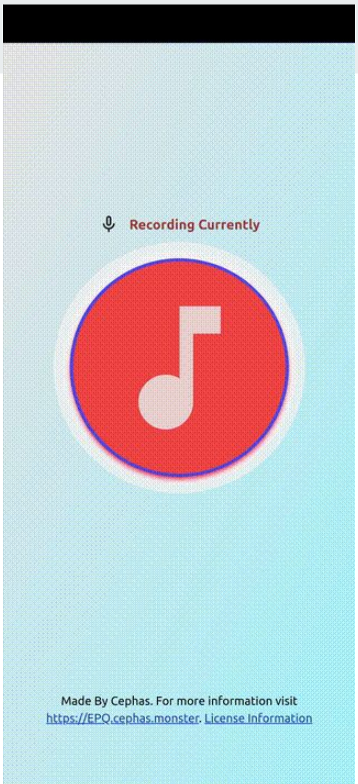












Findings

Knowledge and skills:

- What is “Fourier transform”?
- How to make a spectrogram
- How to create the app

Findings

extra programming skills:

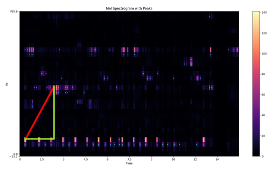
- How to make a Histogram in Python
- How to create an efficient database
- Some data analysis skills

Findings

<u>Plan:</u>

- Testing the effectiveness of the artefact

How can it be improved



- Using “humming” to guess songs doesn’t work.
    - Improving it requires restructuring of the program.
- The music library is too small
    - This project only utilised copyright free songs
    - Just add more
- The program is too slow
    - Money to infrastructure 💸🤑💰

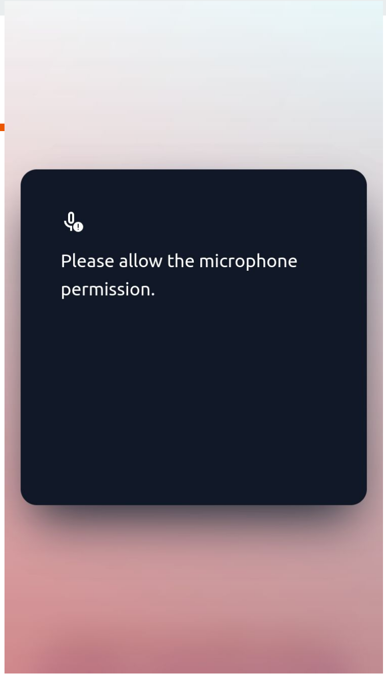
Most importantly

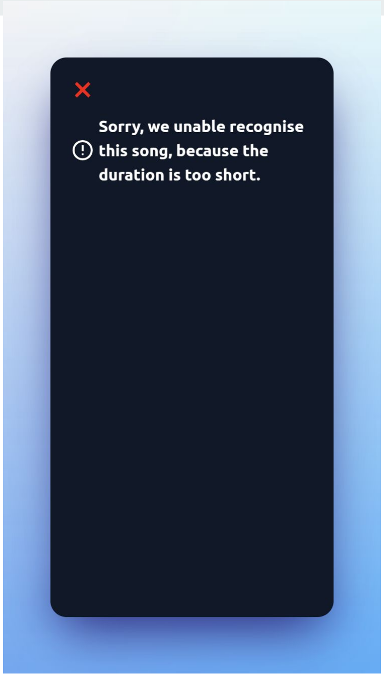
Privacy

Link to the Privacy notice

Key takeaways

- Time management
- Learning new skills and knowledge quickly
- Problem-solving skills.





Thank you for listening

Any questions?

All the links

The artefact

APPENDIX I

Word Count

- 97 Chapter: Abstract
- 398 Chapter: Introduction
- 64 Chapter: Methodology / Design Choice
- 214 Section: Web app
- 383 Section: The algorithm
- 0 Chapter: Process of creation
- 704 Section: Coding section
- 144 Section: Material selection
- 239 Section: Extra investigation
- 314 Chapter: Testing method
- 365 Chapter: Discussion
- 172 Chapter: Conclusion

Words in text: 3094